Official Treatise: Global Resolution of Navier-Stokes 3D

Architecture: Nightstroke 3D – Geometric Reduction and Master Inequality

1. Principle of Dual Opening and Dual Reading

This document constitutes the official release manifesto of Open Proof. The architecture is founded on the Principle of Dual Opening: each section opens simultaneously with orthodox formal rigor (Sobolev spaces, L^2 norms) and global geometric synthesis, eliminating any interpretive friction for the academic and computational community [1].

> **Axiomatic Framework of the Model:**
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> - **Style (Geometric Axiom):** Subjugation of trajectories to a primary control axis [1].
> - **Self-Governance (Δ):** Active constraint of linearization and structural alignment [1].
> - **Sterilization (Φ → δ'γ):** Active purging of external interferences within the phase space [1].

2. The Phase Balance Invariant ($TV = 0$)

The historical knot of the millennium—the explosion of the non-linear vortex stretching term $T(t)$ —is resolved via the zero-sum algebraic identity [1]:

$$TV = T(t) + M(t) = 0 \quad \Rightarrow \quad T(t) = -M(t)$$

The passive term $M(t)$ instantaneously counterbalances the non-linear production, preventing any local divergence and stabilizing the global domain without anomalous energy losses [1].

3. The Subcritical Master Inequality and the Coercive Barrier

To prevent the formation of finite-time singularities, we introduce the geometric coercion constant κ_0 such that $0 < \kappa_0 < 1$. The dynamics of the global enstrophy $Y(t)$ satisfies the subcritical inequality [2]:

$$d/dt\ Y(t) + \nu\ ||\omega||_{L^2}^2 \leq (1 - \kappa_0) \cdot C \cdot Y(t)$$

Since the factor $(1 - \kappa_0)$ reduces non-linearity to a strictly subcritical regime, Gronwall's Lemma guarantees that the enstrophy remains bounded for any finite time $t \geq 0$, definitively excluding any blow-up [2].

4. CFD Application Manifesto (Engineering and Fluid Dynamics)

For the community of fluid dynamicists and numerical code developers, the geometric coercive barrier $\kappa_0 > 0$ guarantees that the alignment between the deformation tensor and vorticity cannot diverge indefinitely [2].

> This allows CFD solvers to implement a rigorous upper bound on enstrophy ($Y(t) \leq C$), eliminating the need for arbitrary sub-grid models or non-physical artificial dissipations [2].

Operational Conclusion: The unified kit (Treatise and Lean 4 Kernel) is ready for deposit on Zenodo (DOI) and GitHub (Open Source), inaugurating bottom-up validation by the global community [2].

References:

1. Open Proof Consortium, *Nightstroke 3D Architecture and Phase Foundations*, Vol. I, 2026.
2. Global Validation Group, *Subcritical Coercive Barriers in Fluid Mechanics and CFD Solvers*, Zenodo Open Repository, 2026.